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AnalystLab Africa - Data Science Internship Programme

Week 4 Assignment

Week 4 Project Summary

**Improving Patient Appointment Attendance and Healthcare Support Using Data and
AI**

HealthConnect Clinic Experience Lab

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Internship: Data Science

Batch: D

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Week 4 Project Summary

Problem addressed

The Data Science track examined whether HealthConnect's historical appointment information can support early prediction of appointment no-shows. The intended value is to help clinic staff prioritise suitable attendance support and improve use of scheduled appointment capacity.

Resources used

- Week 4 assignment brief
- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary
- HealthConnect Clinic Knowledge Base

Key observations

The dataset contains 5,000 appointment records, 18 variables and 1,696 anonymised patient identifiers. Outcomes include 2,423 no-shows, 2,314 attended appointments and 263 cancellations. No exact duplicate rows or duplicate appointment identifiers were found.

- Distance is missing in 90 records and waiting time in 60 records.
- The 1,366 reminder-channel values recorded as None all correspond to no reminder being sent and represent a meaningful category.
- The observed CSV date format differs from the data dictionary's stated ISO format.
- There are 737 Sunday appointments despite the stated Sunday closure.
- Repeated patient identifiers contain inconsistent age histories, limiting longitudinal interpretation.

Proposed approach

The initial analytical task is supervised binary classification. No-Show will be encoded as 1 and Attended as 0. Cancellations will be preserved but temporarily excluded, producing 4,737 eligible records with a 51.15% no-show rate.

Only information available before the prediction time should be used. Logistic regression is proposed as the first interpretable model after a simple baseline. Selected tree-based models may later be compared. Evaluation should include recall, precision, F1-score, confusion matrices, ranking performance, calibration and subgroup error analysis rather than accuracy alone.

Key considerations

The main considerations are the synthetic nature of the data, uncertain timing of reminder and waiting-time variables, cancellation handling, leakage, patient-ID inconsistencies, fairness and the need for a defined staff intervention. Predictive associations must not be interpreted as explanations of patient behaviour.

Proposed Week 5 focus

Week 5 should conduct structured exploratory analysis, investigate the identified quality concerns, confirm the operational prediction point and create a documented modelling cohort with leakage-safe preparation and evaluation design.

Week 4 conclusion: Suitable for an educational prototype; not yet for operational use.
